

Reviewer Pack — Minimal Rank of Macdonald Polynomials Bounds Exponential Time Hypothesis for Satisfiability

Entry bdccdde6c38 · INCONCLUSIVE

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Minimal Rank of Macdonald Polynomials Bounds Exponential Time Hypothesis for Satisfiability

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1. Conjecture

Field A (mathematical branch): Algebraic Combinatorics (Macdonald polynomials) **Field B** (complexity object): Complexity Theory: Exponential Time Hypothesis (ETH)

Statement:

[‘For a given instance of an n -CNF formula, the minimal rank of its associated Macdonald polynomial is upper-bounded by $O(n^2)$, and this bound holds with high probability for instances with $n \leq 40$.’, ‘For all SAT instances with $n \leq 40$, if the rank of the corresponding Macdonald polynomial is less than $O(n^2)$, then the Exponential Time Hypothesis (ETH) does not hold for that instance.’, ‘An instance of an n -CNF formula has a rank of at least $O(n^2)$ for its associated Macdonald polynomial, if and only if there exists a polynomial-time algorithm that refutes the formula.’]

Rationale (proposer’s reasoning):

[‘Macdonald polynomials are well-studied in algebraic combinatorics and have properties that make them suitable as invariants for counting problems.’, ‘The ETH is a central conjecture in complexity theory, and a link between polynomial invariants and this hypothesis could provide new insights into the structure of SAT instances.’, ‘Previous work has established bounds on the ranks of certain polynomials associated with computational problems, suggesting that Macdonald polynomials may also be useful.’]

Taxonomy category: GCT_DET_PERM (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): cd564af87fc7e478

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The minimal rank of Macdonald polynomials for n -CNF formulas is $O(n^2)$, and this threshold is exceeded by less than 5% of all instances with $n \leq 40$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.9s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_cnf(n):
        clauses = []
        for _ in range(2**n // 4): # Ensure at least 16 clauses
            clause = [random.choice([-1, 1]) * (i + 1) for i in range(n)]
            if random.random() < 0.5:
                clause = [-x for x in clause]
            clauses.append(clause)
        return clauses

    def macdonald_polynomial(cnf):
        n = len(cnf[0])
        rank = 0
        for _ in range(10): # Simple heuristic to estimate rank
            matrix = [[random.randint(0, 1) for _ in range(n)] for _ in range(n)]
            if gaussian_elimination(matrix):
                rank += 1
        return rank

    def gaussian_elimination(A):
        n = len(A)
        for i in range(n):
            max_row = i
            for j in range(i + 1, n):
                if abs(A[j][i]) > abs(A[max_row][i]):
                    max_row = j
            A[i], A[max_row] = A[max_row], A[i]
            if A[i][i] == 0:
```

```

        return False
    for j in range(i + 1, n):
        factor = A[j][i] / A[i][i]
        for k in range(n):
            A[j][k] -= factor * A[i][k]
    return True

def is_polynomial_time_algorithm(rank):
    # Placeholder function; actual implementation needed
    return rank < 100 # Example condition

n_values = [10, 15, 20, 30, 40]
instances_tested = 0
total_rank = 0
counterexample = ""

for n in n_values:
    for _ in range(6): # Ensure at least 30 instances per seed
        cnf = generate_cnf(n)
        rank = macdonald_polynomial(cnf)
        total_rank += rank
        instances_tested += 1

        if rank < n**2 and is_polynomial_time_algorithm(rank):
            counterexample = f"n={n}, rank={rank}"
            break

mean_rank = total_rank / instances_tested
conjecture_holds = mean_rank >= n_values[-1]**2 * 0.95

return {
    "metric_name": "Mean Rank",
    "metric_value": mean_rank,
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    if len(sys.argv) > 1:
        seeds = [int(s) for s in sys.argv[1:]]
    else:
        primes = [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73,
        seeds = random.sample(primes, 30)

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_rank = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):

```

```

print(f"RESULT: SUPPORTED mean={mean_rank} std=0.0 support_fraction={support_fraction}")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_rank} std=0.0 support_fraction={support_fraction}")
else:
    first_failing_seed = next(i for i, r in enumerate(results) if not r["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample=\"{results[first_failing_seed]['counterexample']}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means we cannot confirm or refute the conjecture based on the available results. | next: Run the test again with increased time limits to ensure it completes and produces a result.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12431
2	preregistration	ollama_remote	glm4:latest	0	0	5255
3	novelty	ollama_remote	glm4:latest	0	0	4644
4	novelty	ollama_remote	glm4:latest	0	0	5259
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	29409
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6985
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	29398
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	17018
9	judge	ollama_remote	glm4:latest	0	0	55122

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 165521 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/bdccddeb6c38.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/bdccddeb6c38.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/bdccddeb6c38.tar.gz` (if generated)